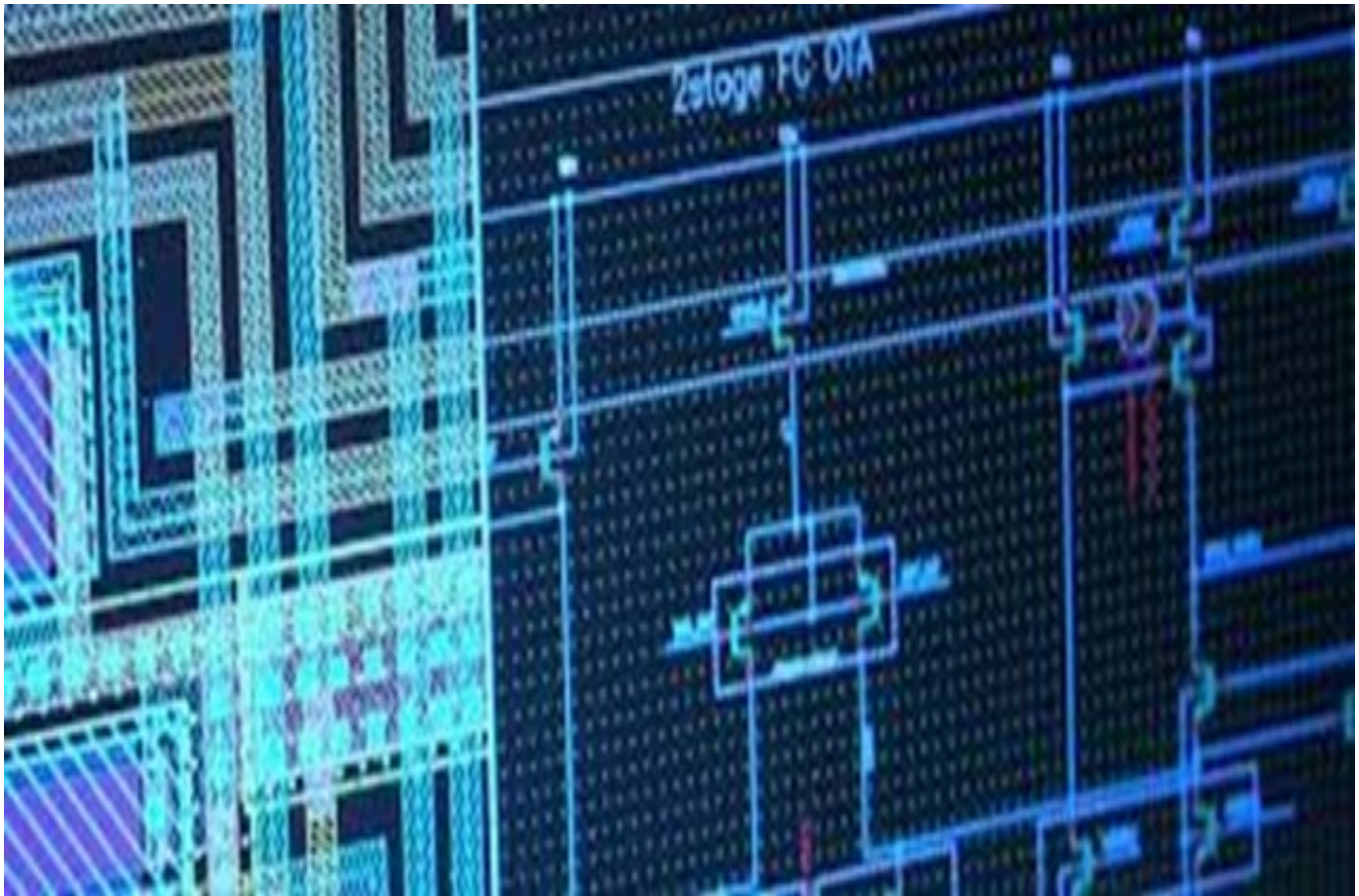




Qualification Pack



Semiconductor IC Design Associate

QP Code: MSU/ELE/Q0101

Version: 1.0

NSQF Level: 4.5

Medhavi Skills University || Medhavi Skills University - Awarding Body Topakhani
Lower Chisopani PO: Singtam Bazar || email:rajat.s@msu.edu.in



Qualification Pack

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Qualification Pack

MSU/ELE/Q0101: Semiconductor IC Design Associate

Brief Job Description

On completion of this qualification the candidate will be able to design and verify basic digital logic gates and combinational circuits. This training imparts knowledge on semiconductor IC design, covering electronics fundamentals, the entire microchip development cycle from conceptualization to packaging, and hands-on experience with EDA tools for design rule check and layout versus schematic verification. Participants will also learn about fabrication steps, design for manufacturability, and optical proximity correction.

Personal Attributes

To excel in Semiconductor IC Design, individuals must possess a strong foundation in electronics and a keen attention to detail for accurate circuit design and verification. A systematic approach, problem-solving abilities, and a curious mindset are essential for navigating the complexities of the microchip development cycle. Proficiency in EDA tools and a strong grasp of fabrication processes are crucial for efficient design and manufacturing. Additionally, a collaborative spirit is valuable for working within design teams and a continuous drive to learn is necessary to stay updated with the evolving semiconductor industry.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [MSU/ELE/N0101: Foundation of Semiconductor Design - Understanding the Basics](#)
2. [MSU/ELE/N0102: Layout Design Development Process- From Theory to Practice](#)
3. [MSU/ELE/N0103: Schematic Design Development Process- From Theory to Practice](#)
4. [MSU/ELE/N0104: Design Rule Check \(DRC\), LVS \(Layout Vs Schematic\) and Reliability Checks](#)
5. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Electronics
Sub-Sector	Strategic Electronics
Occupation	Semiconductor IC Design
Country	India



Qualification Pack

NSQF Level	4.5
Credits	18
Aligned to NCO/ISCO/ISIC Code	NCO-2015/3118.0302, 2152.0502
Minimum Educational Qualification & Experience	Completed 3 year diploma after 10th OR Completed 1st year of UG (UG Certificate) OR Pursuing 2nd year of 2-year diploma after 12th OR Pursuing 3rd year of 3-year diploma after 10th OR Previous relevant Qualification of NSQF Level (4) with 1-2 Years of experience
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	29/05/2027
NSQC Approval Date	30/05/2024
Version	1.0
Reference code on NQR	NG-4.5-EH-02707-2024-V1-MSU
NQR Version	1



Qualification Pack

MSU/ELE/N0101: Foundation of Semiconductor Design - Understanding the Basics

Description

This OS unit is about Semiconductor IC Design. It covers fundamentals of electronics and electronic components which are essential for learning state of the art semiconductor microchip design.

Scope

The scope covers the following :

- Introduction to the world of Semiconductor Microchips
- Electronics for Microchip Design - Fundamental & Concepts
- The Controlled Conductor - Doped Semiconductor
- CMOSFET in VLSI - The game changer
- The language of CMOS - Binary Numbers & Digital Electronics

Elements and Performance Criteria

Introduction to the world of Semiconductor Microchips

To be competent, the user/individual on the job must be able to:

PC1. Understanding the Semiconductor Industry

Electronics for Microchip Design - Fundamental & Concepts

To be competent, the user/individual on the job must be able to:

PC2. The Fundamentals of Electric Current & Voltage

PC3. Electromigration

PC4. Power & Performance of the circuit

PC5. The Capacitors

PC6. Understanding the Time Constant by changing the value of R & C

PC7. Crosstalk by Capacitive Coupling

PC8. Inductors

PC9. AC & DC in Microchips

The Controlled Conductor - Doped Semiconductor

To be competent, the user/individual on the job must be able to:

PC10. Atomic Structure and the free electrons

PC11. Insulator, Conductor & Semiconductor

PC12. Extrinsic Semiconductors

PC13. Diode- Biasing & Working

PC14. Breakdown in a diode- Zenner & Avalanche

PC15. Half wave & Full Wave Rectifier

PC16. Converting Pulsated DC to smooth DC using capacitor

CMOSFET in VLSI - The game changer



Qualification Pack

To be competent, the user/individual on the job must be able to:

PC17. The evolution of a MOSFET - From BJT to JFET to MOSFET

PC18. Structure and Working of a MOSFET

PC19. Electrical Components in MOSFET

PC20. The Short Channel Effects & The Leakage Currents

PC21. Inverter using CMOS

PC22. Power Loss in CMOS

PC23. Latch up in CMOS & Body biasing

The language of CMOS - Binary Numbers & Digital Electronics

To be competent, the user/individual on the job must be able to:

PC24. Number System – Binary, decimal, Octal & Hexadecimal

PC25. The logic Gates

PC26. Designing 4:16 Decoder

PC27. Designing 16:1 Multiplexer

PC28. Creating Circuits from Truth Tables

PC29. Sequential Circuits

PC30. Clocked Circuits

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Facts and challenges associated with semiconductor microchips.

KU2. The semiconductor industry in general.

KU3. Fundamentals of Electric Current & Voltage.

KU4. Foundational principles of Resistance, Capacitance and Inductance as they relate to Microchips.

KU5. Knowledge on why semiconductor material is used for designing microchips

KU6. Fundamentals of semiconductor devices such as diodes and transistors.

KU7. The evolution of CMOS technology followed by the BJT and FET technologies

KU8. The logic Gates, corresponding truth table and Boolean expression

KU9. The combinational circuits like decoder and multiplexers

KU10. The Sequential Circuits like Latches and Flip Flops



Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to the world of Semiconductor Microchips</i>	2	-	-	-
PC1. Understanding the Semiconductor Industry	2	-	-	-
<i>Electronics for Microchip Design - Fundamental & Concepts</i>	26	-	-	-
PC2. The Fundamentals of Electric Current & Voltage	4	-	-	-
PC3. Electromigration	4	-	-	-
PC4. Power & Performance of the circuit	3	-	-	-
PC5. The Capacitors	4	-	-	-
PC6. Understanding the Time Constant by changing the value of R & C	5	-	-	-
PC7. Crosstalk by Capacitive Coupling	3	-	-	-
PC8. Inductors	2	-	-	-
PC9. AC & DC in Microchips	1	-	-	-
<i>The Controlled Conductor - Doped Semiconductor</i>	25	-	-	-
PC10. Atomic Structure and the free electrons	2	-	-	-
PC11. Insulator, Conductor & Semiconductor	3	-	-	-
PC12. Extrinsic Semiconductors	4	-	-	-
PC13. Diode- Biasing & Working	5	-	-	-
PC14. Breakdown in a diode- Zenner & Avalanche	4	-	-	-
PC15. Half wave & Full Wave Rectifier	5	-	-	-
PC16. Converting Pulsated DC to smooth DC using capacitor	2	-	-	-



Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>CMOSFET in VLSI - The game changer</i>	21	-	-	-
PC17. The evolution of a MOSFET - From BJT to JFET to MOSFET	2	-	-	-
PC18. Structure and Working of a MOSFET	5	-	-	-
PC19. Electrical Components in MOSFET	3	-	-	-
PC20. The Short Channel Effects & The Leakage Currents	4	-	-	-
PC21. Inverter using CMOS	3	-	-	-
PC22. Power Loss in CMOS	2	-	-	-
PC23. Latch up in CMOS & Body biasing	2	-	-	-
<i>The language of CMOS - Binary Numbers & Digital Electronics</i>	26	-	-	-
PC24. Number System – Binary, decimal, Octal & Hexadecimal	1	-	-	-
PC25. The logic Gates	5	-	-	-
PC26. Designing 4:16 Decoder	4	-	-	-
PC27. Designing 16:1 Multiplexer	4	-	-	-
PC28. Creating Circuits from Truth Tables	6	-	-	-
PC29. Sequential Circuits	4	-	-	-
PC30. Clocked Circuits	2	-	-	-
NOS Total	100	-	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	MSU/ELE/N0101
NOS Name	Foundation of Semiconductor Design - Understanding the Basics
Sector	Electronics
Sub-Sector	
Occupation	Semiconductor IC Design
NSQF Level	4.5
Credits	2
Version	1.0
Last Reviewed Date	30/05/2024
Next Review Date	29/05/2027
NSQC Clearance Date	30/05/2024



Qualification Pack

MSU/ELE/N0102: Layout Design Development Process- From Theory to Practice

Description

This OS Unit is about Semiconductor IC Design. It covers fundamentals of electronics and electronic components which are essential for learning state of the art semiconductor microchip design. Candidates will learn about Fabrication of CMOS - Lithography at nanoscale, Layout Design Editor - Introduction, and Layout Design Engineering.

Scope

The scope covers the following :

- The Fabrication of CMOS - Lithography at nanoscale
- Layout Design Editor - Introduction
- The Layout Design Engineering - Let's develop the Masks for Lithography

Elements and Performance Criteria

The Fabrication of CMOS - Lithography at nanoscale

To be competent, the user/individual on the job must be able to:

- PC1.** The Fabrication Process
- PC2.** Mask shift Variations
- PC3.** Well Proximity Effect
- PC4.** Antenna Effect and Mitigation techniques
- PC5.** Advanced Lithography Techniques

Layout Design Editor - Introduction

To be competent, the user/individual on the job must be able to:

- PC6.** Layout Design Editor Components
- PC7.** Concept of Layer and Datatypes
- PC8.** Shapes and their Properties
- PC9.** Creating Rectangles in different layers
- PC10.** Understanding the concept of labels
- PC11.** Instantiating a Layout and understanding it's properties
- PC12.** Hierarchies and Nested Instances
- PC13.** Orientation of the device
- PC14.** Changing the Grid and performing stretch Operation
- PC15.** Performing Boolean operations on different Layers
- PC16.** Understanding the key setup files and formats
- PC17.** Exploring objects- Shape, Path, Label, Boundary, Instance, Array

The Layout Design Engineering - Let's develop the Masks for Lithography

To be competent, the user/individual on the job must be able to:



Qualification Pack

- PC18.** Contact, Metal & Vias
- PC19.** Gate Length and Device Width
- PC20.** Layout Design Techniques- Folding, Node Sharing and Matching
- PC21.** Best Layout Design Practices
- PC22.** Standard Cells
- PC23.** Design a Layout for 4:16 Decoder
- PC24.** Design a Layout for 8:1 Multiplexer
- PC25.** Draw a Layout for 2 Input NAND and NOR Gate with P:N ratio as 2:1 and compare the areas
- PC26.** Draw a Layout for 3 input NAND and NOR gate with P:N ratio as 3:1
- PC27.** Draw a Layout of Inverter with the given Specifications and split the gate into 3,2 and 4 folds
- PC28.** In the drawn inverter, identify metal lines more prone to EM
- PC29.** In the pre drawn inverter, provide crosstalk shielding to the output signal.
- PC30.** Make a Layout of a 3-inverter chain by sharing the nodes
- PC31.** Identifying overhang Metals, Missing Vias, Signals prone to cross-talk in the pre drawn layouts

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** The Fabrication steps of a microchip
- KU2.** The challenges involved in fabricating a CMOS device
- KU3.** Understanding of Antenna Effect and its mitigation techniques
- KU4.** Few Advanced level lithography techniques to construct CMOS
- KU5.** The Layout Design Editor Components such as: Canvas, Shapes, Drawing Grid, Layer, Datatypes, Co-ordinates
- KU6.** The key setup files and formats such as: GDSII & OASIS, Layer-map file, Technology File
- KU7.** Standard work practices for designing a Layout Design
- KU8.** Efficient design methodologies and techniques adopted by leading semiconductor design teams
- KU9.** Usage of Layout Editor tool for designing the layout



Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>The Fabrication of CMOS - Lithography at nanoscale</i>	15	-	-	-
PC1. The Fabrication Process	6	-	-	-
PC2. Mask shift Variations	2	-	-	-
PC3. Well Proximity Effect	2	-	-	-
PC4. Antenna Effect and Mitigation techniques	3	-	-	-
PC5. Advanced Lithography Techniques	2	-	-	-
<i>Layout Design Editor - Introduction</i>	5	30	-	-
PC6. Layout Design Editor Components	-	4	-	-
PC7. Concept of Layer and Datatypes	1	8	-	-
PC8. Shapes and their Properties	-	2	-	-
PC9. Creating Rectangles in different layers	-	2	-	-
PC10. Understanding the concept of labels	-	1	-	-
PC11. Instantiating a Layout and understanding it's properties	-	3	-	-
PC12. Hierarchies and Nested Instances	-	5	-	-
PC13. Orientation of the device	-	2	-	-
PC14. Changing the Grid and performing stretch Operation	-	1	-	-
PC15. Performing Boolean operations on different Layers	-	2	-	-
PC16. Understanding the key setup files and formats	3	-	-	-
PC17. Exploring objects- Shape, Path, Label, Boundary, Instance, Array	1	-	-	-
<i>The Layout Design Engineering - Let's develop the Masks for Lithography</i>	10	40	-	-



Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC18. Contact, Metal & Vias	-	2	-	-
PC19. Gate Length and Device Width	1	2	-	-
PC20. Layout Design Techniques- Folding, Node Sharing and Matching	3	2	-	-
PC21. Best Layout Design Practices	5	-	-	-
PC22. Standard Cells	1	2	-	-
PC23. Design a Layout for 4:16 Decoder	-	4	-	-
PC24. Design a Layout for 8:1 Multiplexer	-	4	-	-
PC25. Draw a Layout for 2 Input NAND and NOR Gate with P:N ratio as 2:1 and compare the areas	-	4	-	-
PC26. Draw a Layout for 3 input NAND and NOR gate with P:N ratio as 3:1	-	4	-	-
PC27. Draw a Layout of Inverter with the given Specifications and split the gate into 3,2 and 4 folds	-	4	-	-
PC28. In the drawn inverter, identify metal lines more prone to EM	-	2	-	-
PC29. In the pre drawn inverter, provide crosstalk shielding to the output signal.	-	2	-	-
PC30. Make a Layout of a 3-inverter chain by sharing the nodes	-	4	-	-
PC31. Identifying overhang Metals, Missing Vias, Signals prone to cross-talk in the pre drawn layouts	-	4	-	-
NOS Total	30	70	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	MSU/ELE/N0102
NOS Name	Layout Design Development Process- From Theory to Practice
Sector	Electronics
Sub-Sector	
Occupation	Semiconductor IC Design
NSQF Level	4.5
Credits	5
Version	1.0
Last Reviewed Date	30/05/2024
Next Review Date	29/05/2027
NSQC Clearance Date	30/05/2024



Qualification Pack

MSU/ELE/N0103: Schematic Design Development Process- From Theory to Practice

Description

This OS unit is about Semiconductor IC Design. It covers fundamentals of electronics and electronic components which are essential for learning state of the art semiconductor microchip design. Candidate will learn about schematic design editor and Circuit Design Fundamentals using CMOS Technology.

Scope

The scope covers the following :

- Schematic Design Editor - Introduction
- Circuit Design Fundamentals using CMOS Technology

Elements and Performance Criteria

Schematic Design Editor – Introduction

To be competent, the user/individual on the job must be able to:

PC1. Understanding the Schematic Design Editor Components

Circuit Design Fundamentals using CMOS Technology

To be competent, the user/individual on the job must be able to:

PC2. The Factors Affecting the Circuit Design Functionality & Performance

PC3. Sequential & Clocked Circuits

PC4. Need of the Clocked Circuits

PC5. Inverter Vs Clocked Inverter (Transmission Gate)

PC6. Capacitive & Resistive Load

PC7. Draw a digital circuit for a 4:16 Decoder with Input Ports as A1, A2, A3 and A4 and Output Port as O1-O16

PC8. Export the Circuit created under PC1 and study the schematic Netlist

PC9. Draw a digital circuit for a 16:1 Multiplexer and highlight a net

PC10. Apply a Digital Clock of 5V 60000 Hz to the input of a CMOS Inverter with width of PMOS and NMOS = 4 μ m and length 0.2 μ m

PC11. Observe the transition time and check which is slower 0-1 or 1-0

PC12. Adjust the sizes of PMOS/NMOS such that the output for both type of logic state (logic0 and logic1) takes the equal time to get stable.

PC13. Increase the length of the inverter to 0.6 μ m and observe the change at the output.

PC14. Design an inverter Chain of 5 inverters with applied input A to the first inverter and the output is taken from the 5th inverter as Y. The size of the final inverter is ($w_p = 200\mu$ m, $w_n = 100\mu$ m, $l_n = l_p = 0.2\mu$ m). Tune the sizes of other 4 inverters such that to get the fastest output Y.

PC15. Fold the inverter with size $w_p = 200\mu$ m, $w_n = 100\mu$ m, $l_n = l_p = 0.2\mu$ m into 10,5,4,2 and 1 fold finger. Observe the change at the output.



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- PC16.** Design a 2 input NAND and NOR gate by keeping $w_p = w_n = 10\mu\text{m}$ and $l_n = l_p = 0.2\mu\text{m}$.
- PC17.** • Create an inverter using both the gates. Call it INV_NAND and INV_NOR
• PC18.
- PC18.** Apply 5V 60000 Hz clock at input and check the difference in the outputs.
- PC19.** Tune the sizes of the PMOS and NMOS to get the same characteristic for both the inverters.
- PC20.** Apply VDD (+ve supply 5V) to NMOS ($w=10\mu\text{m}$, $l=0.2\mu\text{m}$). Apply logic 1 at Gate.
- PC21.** Apply VDD (+ve supply 5V) to PMOS ($w=10\mu\text{m}$, $l=0.2\mu\text{m}$). Apply logic 0 at Gate.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** The Schematic Design Editor Components such as: Ports, Nets, Devices
- KU2.** Usage of Schematic Editor tool for designing the layout
- KU3.** To represent electronic circuits using symbols in the Schematic
- KU4.** Principles and Methodologies involved in designing circuits with CMOS (Complementary Metal-Oxide-Semiconductor) technology.
- KU5.** Manipulation of the device properties to understand the functionality
- KU6.** Understand how signals propagate through a circuit
- KU7.** Understand how to tune the devices according to the specifications
- KU8.** The factors affecting the circuit design functionality and performance
- KU9.** The differences between Sequential and Clocked Circuits
- KU10.** Understand CMOS Clocked Inverter (Transmission Gate)



Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Schematic Design Editor - Introduction</i>	5	5	-	-
PC1. Understanding the Schematic Design Editor Components	5	5	-	-
<i>Circuit Design Fundamentals using CMOS Technology</i>	25	65	-	-
PC2. The Factors Affecting the Circuit Design Functionality & Performance	4	-	-	-
PC3. Sequential & Clocked Circuits	4	-	-	-
PC4. Need of the Clocked Circuits	4	-	-	-
PC5. Inverter Vs Clocked Inverter (Transmission Gate)	4	-	-	-
PC6. Capacitive & Resistive Load	2	-	-	-
PC7. Draw a digital circuit for a 4:16 Decoder with Input Ports as A1, A2, A3 and A4 and Output Port as O1-O16	-	5	-	-
PC8. Export the Circuit created under PC1 and study the schematic Netlist	7	1	-	-
PC9. Draw a digital circuit for a 16:1 Multiplexer and highlight a net	-	5	-	-
PC10. Apply a Digital Clock of 5V 60000 Hz to the input of a CMOS Inverter with width of PMOS and NMOS = 4 μ m and length 0.2 μ m	-	5	-	-
PC11. Observe the transition time and check which is slower 0-1 or 1-0	-	3	-	-
PC12. Adjust the sizes of PMOS/NMOS such that the output for both type of logic state (logic0 and logic1) takes the equal time to get stable.	-	4	-	-
PC13. Increase the length of the inverter to 0.6 μ m and observe the change at the output.	-	3	-	-



Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC14. Design an inverter Chain of 5 inverters with applied input A to the first inverter and the output is taken from the 5th inverter as Y. The size of the final inverter is ($w_p = 200\mu\text{m}$, $w_n = 100\mu\text{m}$, $l_n = l_p = 0.2\mu\text{m}$. Tune the sizes of other 4 inverters such that to get the fastest output Y.	-	5	-	-
PC15. Fold the inverter with size $w_p = 200\mu\text{m}$, $w_n = 100\mu\text{m}$, $l_n = l_p = 0.2\mu\text{m}$ into 10,5,4,2 and 1 fold finger. Observe the change at the output.	-	5	-	-
PC16. Design a 2 input NAND and NOR gate by keeping $w_p = w_n = 10\mu\text{m}$ and $l_n = l_p = 0.2\mu\text{m}$.	-	5	-	-
PC17. • Create an inverter using both the gates. Call it INV_NAND and INV_NOR • PC18.	-	10	-	-
PC18. Apply 5V 60000 Hz clock at input and check the difference in the outputs.	-	3	-	-
PC19. Tune the sizes of the PMOS and NMOS to get the same characteristic for both the inverters.	-	5	-	-
PC20. Apply VDD (+ve supply 5V) to NMOS ($w = 10\mu\text{m}$, $l = 0.2\mu\text{m}$). Apply logic 1 at Gate.	-	3	-	-
PC21. Apply VDD (+ve supply 5V) to PMOS ($w = 10\mu\text{m}$, $l = 0.2\mu\text{m}$). Apply logic 0 at Gate.	-	3	-	-
NOS Total	30	70	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	MSU/ELE/N0103
NOS Name	Schematic Design Development Process- From Theory to Practice
Sector	Electronics
Sub-Sector	
Occupation	Semiconductor IC Design
NSQF Level	4.5
Credits	4
Version	1.0
Last Reviewed Date	30/05/2024
Next Review Date	29/05/2027
NSQC Clearance Date	30/05/2024



Qualification Pack

MSU/ELE/N0104: Design Rule Check (DRC), LVS (Layout Vs Schematic) and Reliability Checks

Description

This OS unit is about "Semiconductor IC Design". It covers fundamentals of electronics and electronic components which are essential for learning state of the art semiconductor microchip design. In this OS candidate will learn about Design Rule Check (DRC), LVS (Layout Vs Schematic) and Reliability Checks.

Scope

The scope covers the following :

- Design Rule Check (DRC), LVS (Layout Vs Schematic) and Reliability Checks

Elements and Performance Criteria

Physical Design DRC (Design Rule Check)

To be competent, the user/individual on the job must be able to:

- PC1.** Need of DRC
- PC2.** Inputs Required to verify the DRC
- PC3.** The Design Rule Manual
- PC4.** Basic Rules for layout design development
- PC5.** Verify DRC in pre-drawn Layout Design and observe the DRC by highlighting on the canvas.
- PC6.** Correct DRC in pre-drawn Layout Designs
- PC7.** Develop a Layout and clean the following DRC types: Min Width, Min Spacing, Min Enclosure, Square Via & Contact, Min Extension, Min Area, Off-Grid
- PC8.** Check and Correct DRC in all the layouts drawn under the NOS- Layout Design and Development Process

Physical Design LVS (Layout VS Schematic)

To be competent, the user/individual on the job must be able to:

- PC9.** What is LVS and how does it work
- PC10.** The netlist
- PC11.** Inputs Required to verify the LVS
- PC12.** Types of Errors in LVS
- PC13.** Identify LVS errors in the pre-drawn layouts
- PC14.** Check and correct LVS in all the layouts drawn under the NOS- Layout Design and Development Process

Reliability Checks - IR, EM, ERC, DFM

To be competent, the user/individual on the job must be able to:

- PC15.** Understanding Electromigration
- PC16.** IR Drop & Ground Bounce
- PC17.** Types of ERC



Qualification Pack

PC18. Types of DFM

The ultimate journey of semiconductor microchip Development - Concept to Realization on Silicon

To be competent, the user/individual on the job must be able to:

PC19. PPA - Performance Power Area

PC20. • Most Common Foundational IPs:

- a. IOs
- b. MEMORIES
- c. Analog Designs
- d. SERDES
- e. Standard Cells

PC21. Foundational IP Development Flow

PC22. Placing and connecting the Foundational IPs together to develop the full SOC (System On Chip)

PC23. The PDK

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Importance of the DRC

KU2. The Inputs required to verify the DRC

KU3. Design Rule Checks specific to semiconductor fabrication processes.

KU4. Different Physical design tools to efficiently perform DRC and LVS tasks

KU5. To identify design rule violations and anomalies in layout designs

KU6. To learn techniques for resolving DRC and LVS violations

KU7. To manipulate a DRM for writing a custom rule

KU8. About the Layout and Schematic Netlist

KU9. To identify LVS violations and anomalies

KU10. To learn to visualize/highlight DRC and LVS errors

KU11. To have a broader understanding of reliability checks

KU12. To have an understanding of the Foundational IPs

KU13. To understand the full cycle of a microchip product from concept to functional siliconization.



Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Physical Design DRC (Design Rule Check)</i>	8	40	-	-
PC1. Need of DRC	1	-	-	-
PC2. Inputs Required to verify the DRC	2	-	-	-
PC3. The Design Rule Manual	1	-	-	-
PC4. Basic Rules for layout design development	4	-	-	-
PC5. Verify DRC in pre-drawn Layout Design and observe the DRC by highlighting on the canvas.	-	5	-	-
PC6. Correct DRC in pre-drawn Layout Designs	-	5	-	-
PC7. Develop a Layout and clean the following DRC types: Min Width, Min Spacing, Min Enclosure, Square Via & Contact, Min Extension, Min Area, Off-Grid	-	5	-	-
PC8. Check and Correct DRC in all the layouts drawn under the NOS- Layout Design and Development Process	-	25	-	-
<i>Physical Design LVS (Layout VS Schematic)</i>	13	25	-	-
PC9. What is LVS and how does it work	3	-	-	-
PC10. The netlist	4	-	-	-
PC11. Inputs Required to verify the LVS	2	-	-	-
PC12. Types of Errors in LVS	4	-	-	-
PC13. Identify LVS errors in the pre-drawn layouts	-	5	-	-
PC14. Check and correct LVS in all the layouts drawn under the NOS- Layout Design and Development Process	-	20	-	-
<i>Reliability Checks - IR, EM, ERC, DFM</i>	6	-	-	-
PC15. Understanding Electromigration	1	-	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC16. IR Drop & Ground Bounce	1	-	-	-
PC17. Types of ERC	2	-	-	-
PC18. Types of DFM	2	-	-	-
<i>The ultimate journey of semiconductor microchip Development - Concept to Realization on Silicon</i>	8	-	-	-
PC19. PPA - Performance Power Area	1	-	-	-
PC20. • Most Common Foundational IPs: • a. IOs • b. MEMORIES • c. Analog Designs • d. SERDES • e. Standard Cells	2	-	-	-
PC21. Foundational IP Development Flow	2	-	-	-
PC22. Placing and connecting the Foundational IPs together to develop the full SOC (System On Chip)	2	-	-	-
PC23. The PDK	1	-	-	-
NOS Total	35	65	-	-



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National Occupational Standards (NOS) Parameters

NOS Code	MSU/ELE/N0104
NOS Name	Design Rule Check (DRC), LVS (Layout Vs Schematic) and Reliability Checks
Sector	Electronics
Sub-Sector	
Occupation	Semiconductor IC Design
NSQF Level	4.5
Credits	5
Version	1.0
Last Reviewed Date	30/05/2024
Next Review Date	29/05/2027
NSQC Clearance Date	30/05/2024



Qualification Pack

DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:



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- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e- mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.



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PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings



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- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-



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National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	12/03/2026
Next Review Date	12/03/2029
NSQC Clearance Date	12/03/2026

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Assessment System Overview:

- Batches assigned to the assessment agencies for conducting the assessment on SID or email
- Assessment agencies send the assessment confirmation to VTP/TC looping Madhavi Skills University (MSU)
- Assessment agency deploys the ToA-certified Assessor for executing the assessment
- MSU monitors the assessment process & records

Mode of assessment: - Computerised online assessment theory

- Assessor based practical assessment.

2. Testing Environment:



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- Check the Assessment location, date and time
- If the batch size is more than 30 (For a classroom based assessment), then there should be 2 Assessors.
- Check that the allotted time for the candidates to complete Theory & Practical Assessment is correct.
- Computer with internet connectivity

3. Assessment Quality Assurance levels / Framework:

- Question bank is created by the Subject Matter Experts (SME) are verified by the other SMEs
- Questions are mapped to the specified assessment criteria
- Assessor must be ToA certified & trainer must be ToT Certified
- Minimum 50% passing marks

4. Types of evidence or evidence-gathering protocol:

- Time-stamped & geotagged reporting of the candidate from assessment location
- Centre photographs with signboards and scheme-specific branding

5. Method of verification or validation:

- Computerised online randomised objective type questionnaire for theory
- Surprise visit to the assessment location

6. Method for assessment documentation, archiving, and access

- Soft Copy results are emailed automatically to the concerned authorities and the candidates.
- Hard and copies of the documents are stored

Minimum Aggregate Passing % at QP Level : 50

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to



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successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
MSU/ELE/N0101.Foundation of Semiconductor Design - Understanding the Basics	100	0	0	0	100	10
MSU/ELE/N0102.Layout Design Development Process- From Theory to Practice	30	70	0	0	100	30
MSU/ELE/N0103.Schematic Design Development Process- From Theory to Practice	30	70	0	0	100	25
MSU/ELE/N0104.Design Rule Check (DRC), LVS (Layout Vs Schematic) and Reliability Checks	35	65	0	0	100	25
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	10
Total	215	235	-	-	450	100



Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training



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Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.



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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.